Matteo De Liquori

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Education

ETH Zürich, *Zürich* 2023 – 2027 (expected)

Bachelor Mechanical Engineering

Kantonsschule Wettingen, *Wettingen* 2018 - 2023

Matura *(Focus-Subjects Physics & Mathematics)*

Experience & Relevant Coursework

Fluid Dynamics, *ETH Zürich* Spring Semester 2026

- Introduction to viscous and inviscid flows, boundary layers, laminar and turbulent flows.

Introduction to Finite Element Analysis (FEA), *ETH Zürich* Spring Semester 2026

- Developed custom elementary FEA solvers in Python for various analysis types
- Identification and classification of typical FEA error sources
- Introductory modal analysis in Abaqus CAE (guided coursework)

Machine Learning, *ETH Zürich* May - June 2026

- Implementation of a KNN regressor for obstacle distance estimation from RGB images (robot dataset, 0.12m MAE)
- Implementation of a UNet architecture for binary image segmentation (90% IoU)

Helbling Physical AI Hackathon, *ETH Zürich* 05.05.2026

- Participated in a computer vision challenge to estimate sewer camera position from video
- Contributed to a team solution using AI-assisted development (LLM-based coding)
- Placed 4th out of 16 teams

Innovationsprojekt 2025, *ETH Zürich* Fall Semester 2025

- Development of a small robot on a pre-made self-driving chassis in a 4-person team
- Design of custom components in Fusion 360, manufactured via FDM 3D printing
- Integration of servos and similar components and writing basic Arduino code

Intern, *Paul-Scherrer-Institute* June 2022

- Assembly and disassembly of neutron guide tubes for the UCN experiment setup
- Coating thickness measurement of samples using a contact profilometer
- Data analysis in Excel to determine the optimal coating thickness

Skills & Interests

Software: MSOffice Suite (Word, Excel, PowerPoint etc..), Python (intermediate), C++ (basic), Fusion 360, Arduino, basics in VSCode and GitHub

Hardware: FDM 3D Printing, Mechanical assembly & disassembly

Languages: German & Italian *(Native)*, English *(Fluent)*, French *(Basic)*

Interests: 3D-Printing, Videogames, Aviation, Formula-1, Basketball, Weightlifting